

TRABAJO DE INGENIERÍA DEL CONOCIMIENTO

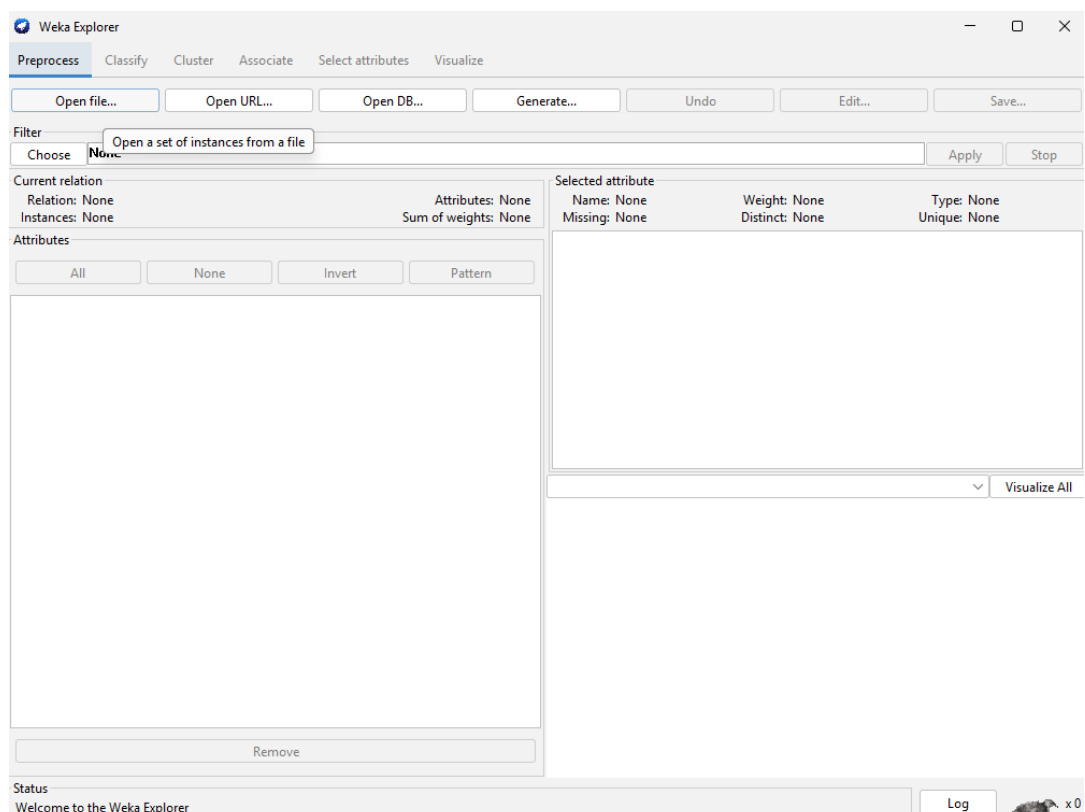
Alumno: Briceño Alaya Yimar Jesus

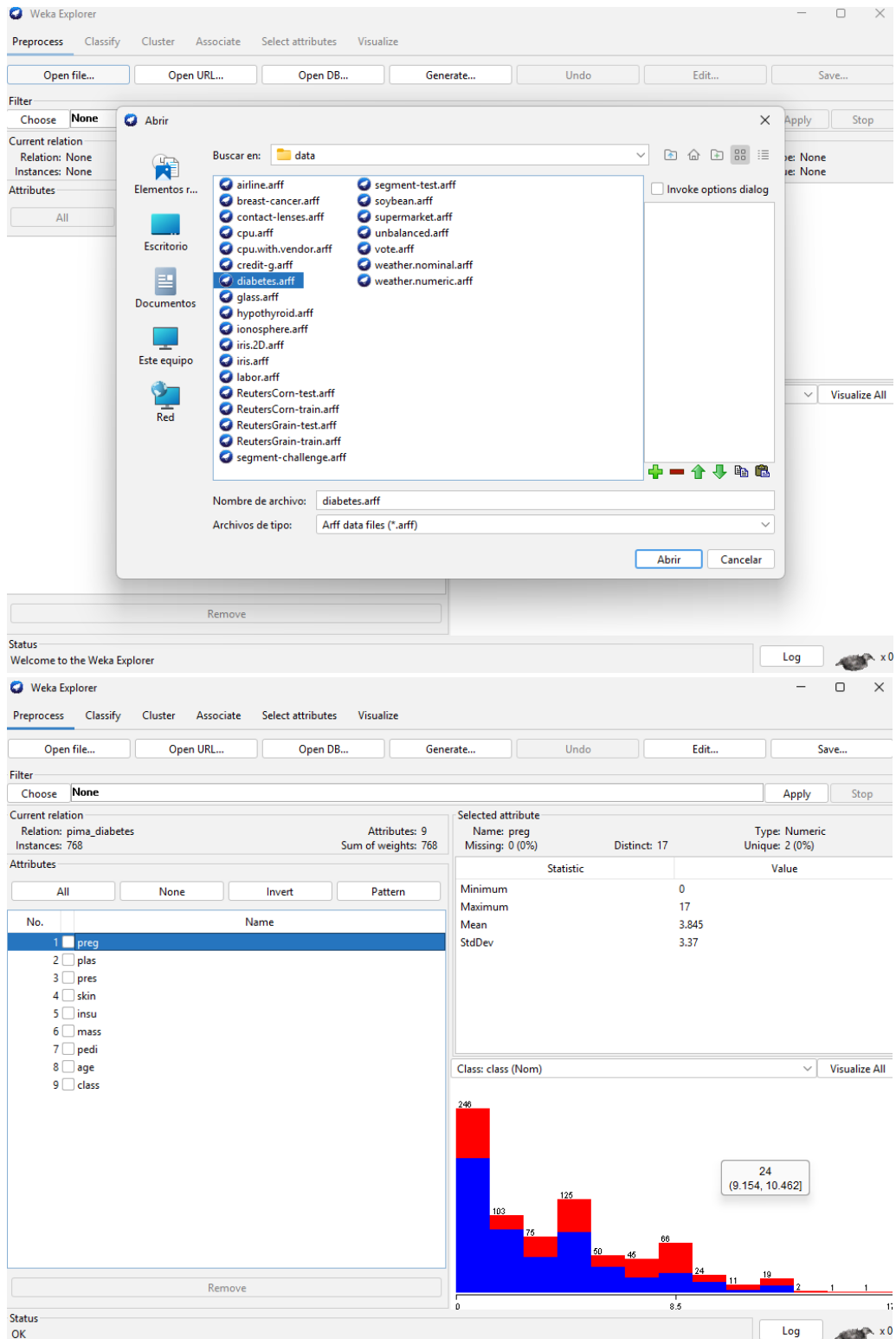
Fecha: 25 de septiembre de 2026

El desarrollo de esta actividad lo haremos mediante pasos secuenciales, empezando desde la carga del dataset que en este caso será la “Diabetes”.

• PASO 1:

Abrimos WEKA y hacemos click en “**Explorer**”. Una vez abierto, en la pestaña superior le damos click a “**Open File**” y buscamos el dataset que viene por defecto con el programa (diabetes.arff), obteniendo la vista previa de los datos en la pestaña *Preprocess*.



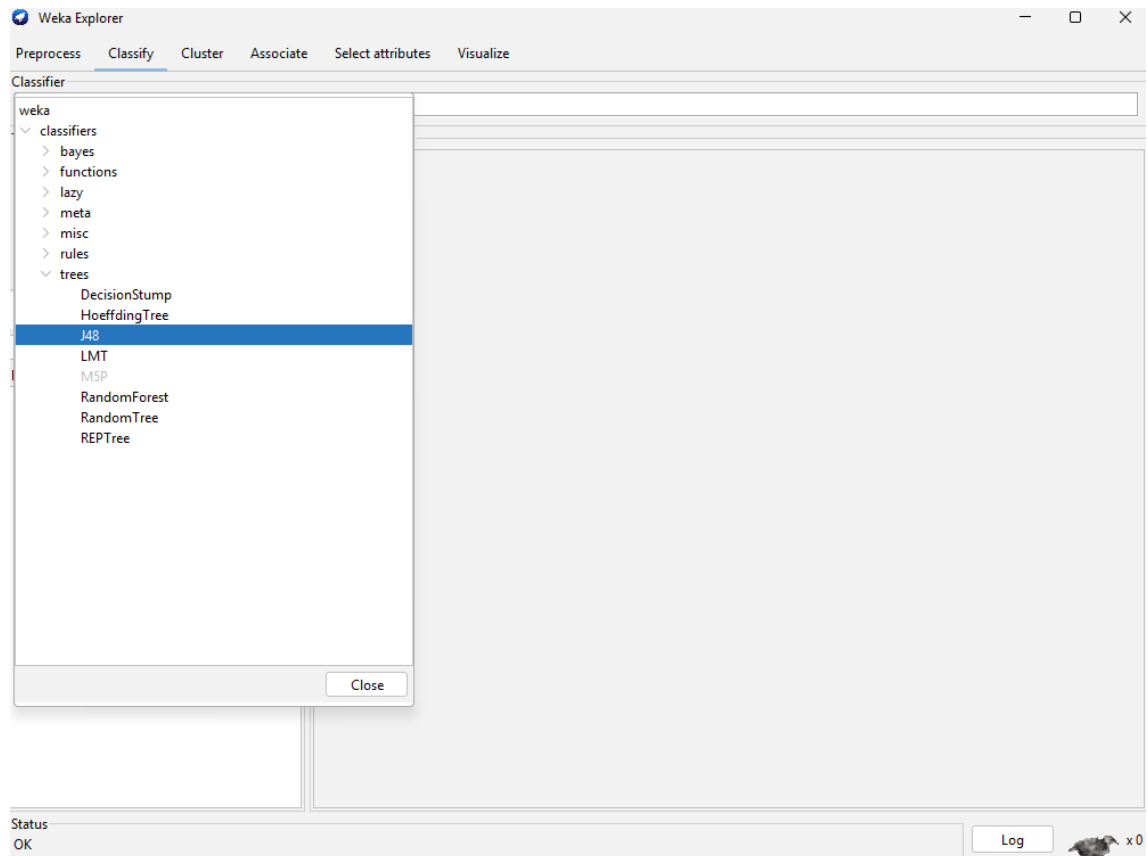


• PASO 2:

Podemos observar que tenemos **768 instancias** y **9 atributos** (preg, plas, pres, skin, insu, mass, pedi, age, class), todos sin filtros aplicados.

Hacemos click en Classify → Choose → trees → J48. En *Test Options*, dejamos

marcado **Cross-validation** con **Folds: 10** y finalmente presionamos en **Start**.



=== Run information ===

```
Scheme:      weka.classifiers.trees.J48 -C 0.25 -M 2
Relation:    pima_diabetes
Instances:   768
Attributes:  9
              preg
              plas
              pres
              skin
              insu
              mass
              pedi
              age
              class
Test mode:   10-fold cross-validation
```

```
=== Classifier model (full training set) ===
```

```
J48 pruned tree
```

```
-----
```

```
plas <= 127
```

```
| mass <= 26.4: tested_negative (132.0/3.0)
| mass > 26.4
| | age <= 28: tested_negative (180.0/22.0)
| | age > 28
| | | plas <= 99: tested_negative (55.0/10.0)
| | | plas > 99
| | | | pedi <= 0.56: tested_negative (84.0/34.0)
| | | | pedi > 0.56
| | | | | preg <= 6
| | | | | age <= 30: tested_positive (4.0)
| | | | | age > 30
| | | | | age <= 34: tested_negative (7.0/1.0)
| | | | | age > 34
| | | | | mass <= 33.1: tested_positive (6.0)
| | | | | mass > 33.1: tested_negative (4.0/1.0)
| | | | | preg > 6: tested_positive (13.0)
```

```
plas > 127
```

```
| mass <= 29.9
| | plas <= 145: tested_negative (41.0/6.0)
| | plas > 145
| | | age <= 25: tested_negative (4.0)
| | | age > 25
| | | | age <= 61
| | | | mass <= 27.1: tested_positive (12.0/1.0)
| | | | mass > 27.1
| | | | | pres <= 82
| | | | | pedi <= 0.396: tested_positive (8.0/1.0)
| | | | | pedi > 0.396: tested_negative (3.0)
| | | | | pres > 82: tested_negative (4.0)
| | | | age > 61: tested_negative (4.0)
| mass > 29.9
| | plas <= 157
| | | pres <= 61: tested_positive (15.0/1.0)
| | | pres > 61
| | | | age <= 30: tested_negative (40.0/13.0)
| | | | age > 30: tested_positive (60.0/17.0)
| | plas > 157: tested_positive (92.0/12.0)
```

```

Number of Leaves :    20

Size of the tree :    39

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances      567           73.8281 %
Incorrectly Classified Instances    201           26.1719 %
Kappa statistic                     0.4164
Mean absolute error                 0.3158
Root mean squared error             0.4463
Relative absolute error             69.4841 %
Root relative squared error         93.6293 %
Total Number of Instances          768

=== Detailed Accuracy By Class ===

                TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                0.814    0.403    0.790     0.814    0.802      0.417    0.751    0.811    tested_neg
                0.597    0.186    0.632     0.597    0.614      0.417    0.751    0.572    tested_pos
Weighted Avg.   0.738    0.327    0.735     0.738    0.736      0.417    0.751    0.727

=== Confusion Matrix ===
|
|  a   b   <-- classified as
| 407  93 |  a = tested_negative
| 108 160 |  b = tested_positive

```

Resumen del resultado inicial:

- **Accuracy:** 73.83% (567/768 correctas)
- **Kappa:** 0.4164 (concordancia moderada)
- **Árbol:** 39 nodos, 20 hojas
- **Matriz de confusión:** De los que realmente tenían diabetes (268 positivos), el modelo solo detectó 160 (*recall* 0.597). Se escaparon 108 casos como falsos negativos, un punto débil que buscaremos mejorar al tratar los datos ruidosos.

• PASO 3:

Visualizamos el árbol que generó el programa haciendo click derecho sobre el resultado en la lista lateral y seleccionando "**Visualize tree**".

Supplied test set: ☐ Cross-validation Folds: 10
☐ Percentage split %: 66
 More options...

(Nom) class: ▼

Start Stop

Result list (right-click for options):

08:46:32 - trees.J48

- View in main window
- View in separate window
- Save result buffer
- Delete result buffer(s)
- Load model
- Save model
- Re-evaluate model on current test set
- Re-apply this model's configuration
- Visualize classifier errors
- Visualize tree
- Visualize margin curve
- Visualize threshold curve
- Cost/Benefit analysis
- Visualize cost curve

Size of the tree : 39

Time taken to build model: 0

=== Stratified cross-validation
 === Summary ===

Correctly Classified Instance
 Incorrectly Classified Instance

Kappa statistic

error
 red error
 ute error
 squared error
 f Instances

Accuracy By Class

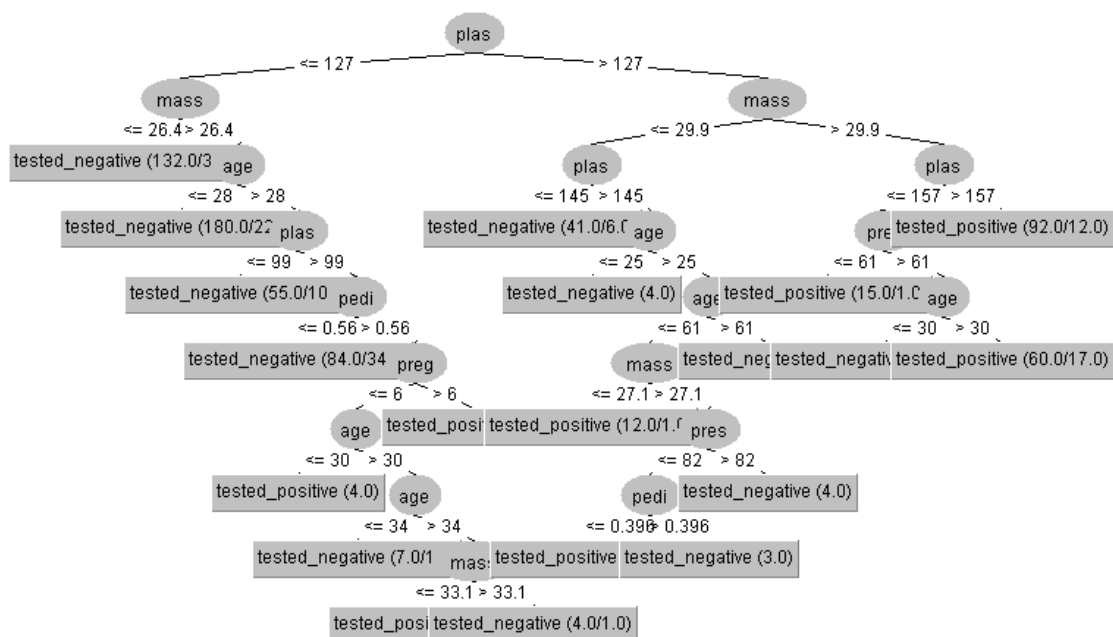
TP Rate	FP
0.814	0.4
0.597	0.1
0.738	0.3

Matrix ===

classified as
 = tested_negat
 = tested_posit

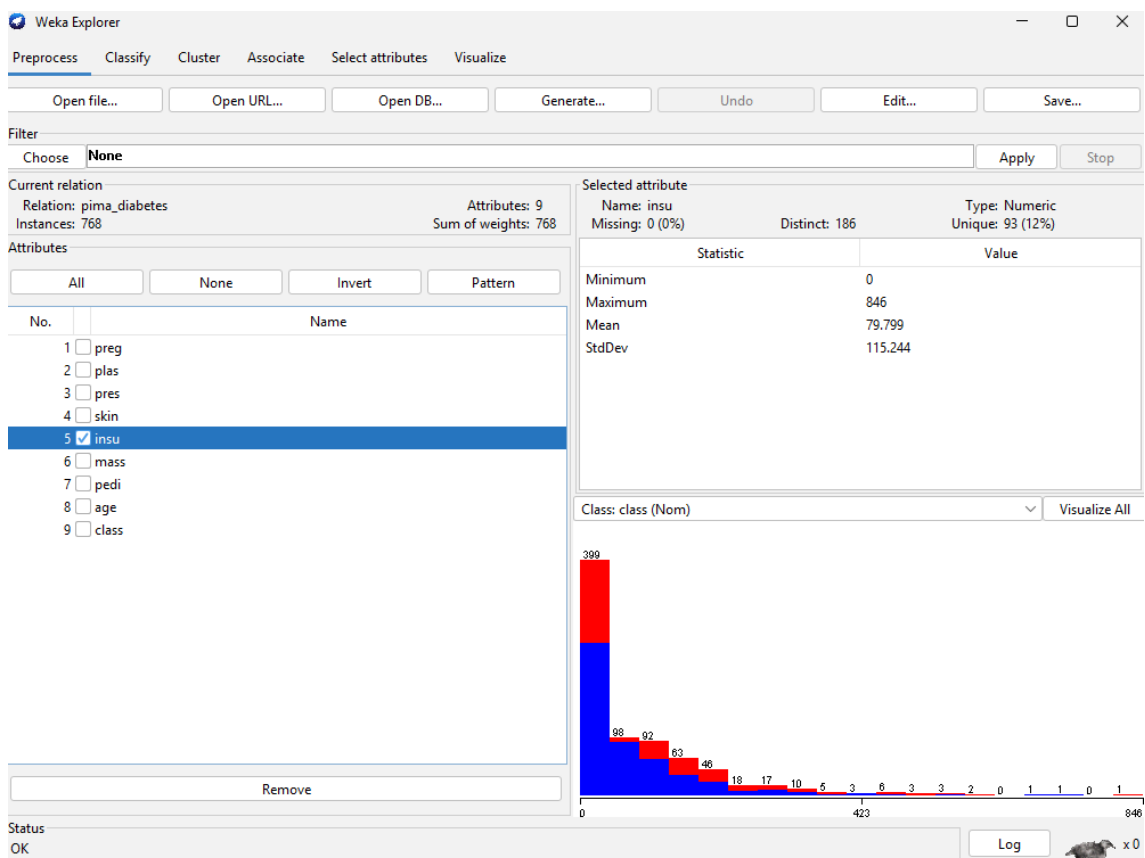
Weka Classifier Tree Visualizer: 08:46:32 - trees.J48 (pima_diabetes)

Tree View



• **PASO 4:** Para el filtrado de información, primero identificamos las variables y su significado:

- preg → cantidad de embarazos (*0 es válido*)
- plas → concentración de glucosa en plasma (*0 es imposible*)
- pres → presión arterial diastólica (*0 es imposible*)
- skin → grosor del pliegue cutáneo del tríceps (*0 es imposible, pero hay muchos*)
- insu → insulina sérica a las 2 horas (*0 es imposible, pero hay muchos*)
- mass → índice de masa corporal / IMC (*0 es imposible*)
- pedi → función de pedigrí de diabetes (*antecedente genético*)
- age → edad (*0 es imposible*)
- class → resultado: positivo o negativo para diabetes



Nota: Observamos que casi 400 datos de la variable insu están en 0. Si los eliminamos todos, nos quedaríamos sin datos suficientes para entrenar el modelo.

- **PASO 5:** Hacemos click en el botón “Edit” (junto a la primera instancia) para revisar y limpiar manualmente los valores imposibles.

The screenshot shows the Orange3 software interface. The top toolbar includes buttons for 'Open file...', 'Open URL...', 'Open DB...', 'Generate...', 'Undo', 'Edit...', and 'Save...'. The 'Edit...' button is highlighted. Below the toolbar, the 'Viewer' window is open, displaying a table of data for the 'pima_diabetes' dataset. The table has 10 columns: 'No.', '1: preg', '2: plas', '3: pres', '4: skin', '5: insu', '6: mass', '7: pedi', '8: age', and '9: class'. The 'class' column contains values like 'tested...'. At the bottom of the window, there are buttons for 'Add instance', 'Undo', 'OK', and 'Cancel'. The status bar at the very bottom shows 'Status OK' and a 'Log' button.

No.	1: preg	2: plas	3: pres	4: skin	5: insu	6: mass	7: pedi	8: age	9: class
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	tested...
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	tested...
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	tested...
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	tested...
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	tested...
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	tested...
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	tested...
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	tested...
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	tested...
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	tested...
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	tested...
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	tested...
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	tested...
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	tested...
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	tested...
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	tested...
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	tested...
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	tested...
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	tested...
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	tested...
21	3.0	126.0	88.0	41.0	235.0	39.3	0.704	27.0	tested...
22	8.0	99.0	84.0	0.0	0.0	35.4	0.388	50.0	tested...
23	7.0	196.0	90.0	0.0	0.0	39.8	0.451	41.0	tested...
24	9.0	119.0	80.0	35.0	0.0	29.0	0.263	29.0	tested...

• PASO 6:

Seleccionamos y eliminamos las instancias con valores fisiológicamente imposibles, siendo conservadores para no perder poder estadístico:

1. Con Pres:

Viewer

Relation: pima_diabetes

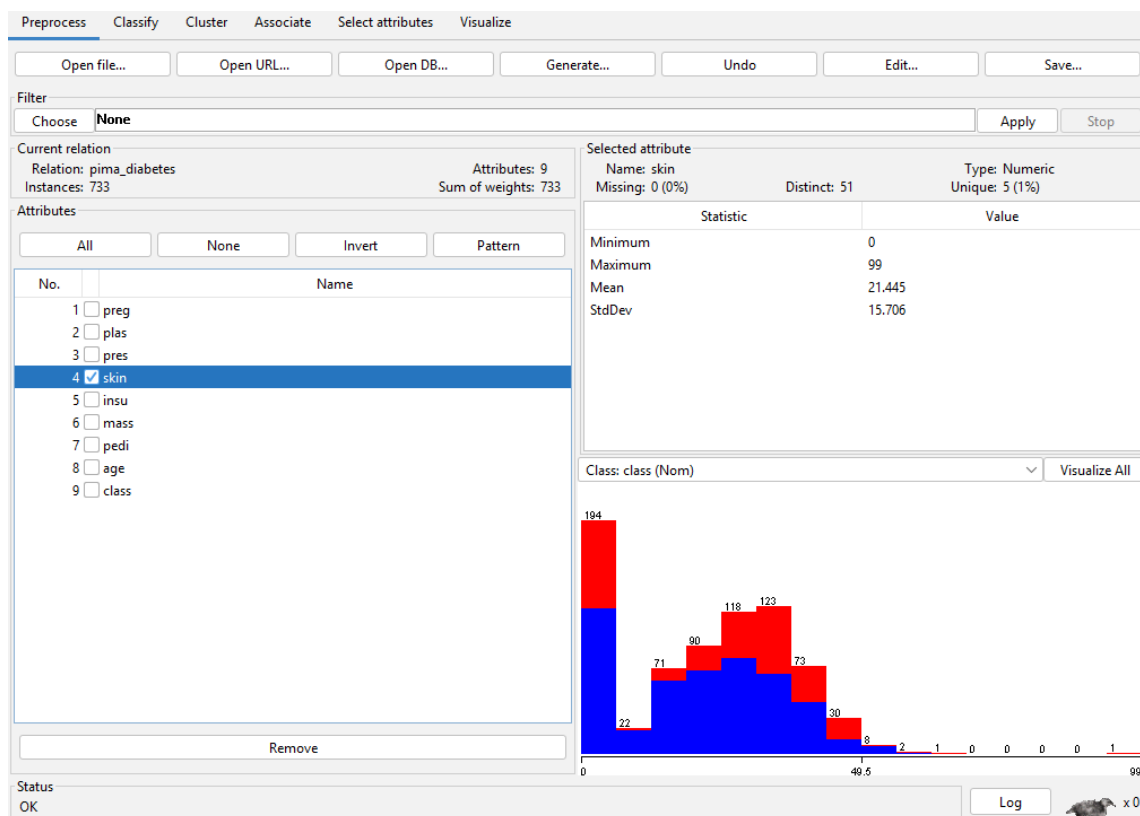
No. 1: preg 2: plas 3: pres 4: skin 5: insu 6: mass 7: pedi 8: age 9: class

Sort view: left click = ascending / Shift + left click = descending
Menu: right click (or left+alt)

17	2.0	129.0	0.0	0.0	0.0	30.0	0.304	41.0	tested...
18	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	tested...
19	0.0	131.0	0.0	0.0	0.0	43.2	0.27	26.0	tested...
20	11.0	135.0	0.0	0.0	0.0	52.3	0.578	40.0	tested...
21	3.0	141.0	0.0	0.0	0.0	30.0	0.761	27.0	tested...
22	0.0	138.0	0.0	0.0	0.0	36.3	0.933	25.0	tested...
23	2.0	146.0	0.0	0.0	0.0	27.5	0.24	28.0	tested...
24	0.0	167.0	0.0	0.0	0.0	32.3	0.839	30.0	tested...
25	1.0	180.0	0.0	0.0	0.0	43.3	0.282	41.0	tested...
26	0.0	141.0	0.0	0.0	0.0	42.4	0.205	29.0	tested...
27	8.0	120.0	0.0	0.0	0.0	30.0	0.183	38.0	tested...
28	0.0	145.0	0.0	0.0	0.0	44.2	0.63	31.0	tested...
29	4.0	132.0	0.0	0.0	0.0	32.9	0.302	23.0	tested...
30	4.0	183.0	0.0	0.0	0.0	28.4	0.212	36.0	tested...
31	0.0	119.0	0.0	0.0	0.0	32.4	0.141	24.0	tested...
32	10.0	115.0	0.0	0.0	0.0	0.0	0.261	30.0	tested...
33	2.0	87.0	0.0	23.0	0.0	28.9	0.773	25.0	tested...
34	13.0	129.0	0.0	30.0	0.0	39.9	0.569	44.0	tested...
35	1.0	89.0	24.0	19.0	25.0	27.8	0.559	21.0	tested...
36	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	tested...
37	1.0	88.0	30.0	42.0	99.0	55.0	0.496	26.0	tested...
38	1.0	109.0	38.0	18.0	120.0	23.1	0.407	26.0	tested...
39	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	tested...
40									

Add instance Undo OK Cancel

Se borraron 35 instancias por pres = 0 (Esto porque no existe persona que tenga presión 0)



2. **Con plas:** Son 194 instancias que se deberían de borrar al igual que insu, pero para conservar el modelo del poder estadístico. (Lo mantendremos por ahora así)

3. **Con mass:**

Viewer

Relation: pima_diabetes

No.	1: preg	2: plas	3: pres	4: skin	5: insu	6: mass	7: pedi	8: age	9: class
	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Nominal
1	0.0	118.0	64.0	23.0	89.0	0.0	1.731	21.0	tested...
2	0.0	102.0	75.0	23.0	0.0	0.0	0.572	21.0	tested...
3	5.0	136.0	82.0	0.0	0.0	0.0	0.64	69.0	tested...
4	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	tested...
5	1.0	97.0	64.0	19.0	82.0	18.2	0.299	21.0	tested...
6	1.0	83.0	68.0	0.0	0.0				
7	1.0	97.0	70.0	15.0	0.0	18.2	0.147	21.0	tested...
8	0.0	104.0	76.0	0.0	0.0	18.4	0.582	27.0	tested...
9	1.0	80.0	55.0	0.0	0.0	19.1	0.258	21.0	tested...
10	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	tested...
11	1.0	103.0	80.0	11.0	82.0	19.4	0.491	22.0	tested...
12	1.0	92.0	62.0	25.0	41.0	19.5	0.482	25.0	tested...
13	1.0	100.0	74.0	12.0	46.0	19.5	0.149	28.0	tested...
14	1.0	95.0	66.0	13.0	38.0	19.6	0.334	25.0	tested...
15	6.0	129.0	90.0	7.0	326.0	19.6	0.582	60.0	tested...
16	6.0	92.0	92.0	0.0	0.0	19.9	0.188	28.0	tested...
17	0.0	105.0	68.0	22.0	0.0	20.0	0.236	22.0	tested...
18	2.0	68.0	62.0	13.0	15.0	20.1	0.257	23.0	tested...
19	1.0	71.0	48.0	18.0	76.0	20.4	0.323	22.0	tested...
20	2.0	99.0	70.0	16.0	44.0	20.4	0.235	27.0	tested...
21	4.0	96.0	56.0	17.0	49.0	20.8	0.34	26.0	tested...
22	9.0	120.0	72.0	22.0	56.0	20.8	0.733	48.0	tested...
23	0.0	101.0	64.0	17.0	0.0	21.0	0.252	21.0	tested...
24	3.0	150.0	76.0	0.0	0.0	21.0	0.207	37.0	tested...

Right click (or left+alt) for context menu

Add instance Undo OK Cancel

Se eliminan 4 instancias.

4. Con plas:

Viewer

Relation: pima_diabetes

No.	1: preg	2: plas	3: pres	4: skin	5: insu	6: mass	7: pedi	8: age	9: class
	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Numeric	Nominal
1	1.0	0.0	48.0	20.0	0.0	24.7	0.14	22.0	tested...
2	1.0	0.0	74.0	20.0	23.0	27.7			
3	1.0	0.0	68.0	35.0	0.0	32.0	0.389	22.0	tested...
4	6.0	0.0	68.0	41.0	0.0	0.0	0.727	41.0	tested...
5	5.0	0.0	80.0	32.0	0.0	41.0	0.346	37.0	tested...
6	5.0	44.0	62.0	0.0	0.0	25.0	0.587	36.0	tested...
7	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	tested...
8	0.0	57.0	60.0	0.0	0.0	21.7	0.735	67.0	tested...
9	9.0	57.0	80.0	37.0	0.0	32.8	0.096	41.0	tested...
10	3.0	61.0	82.0	28.0	0.0	34.4	0.243	46.0	tested...
11	7.0	62.0	78.0	0.0	0.0	32.6	0.391	41.0	tested...
12	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	tested...
13	0.0	67.0	76.0	0.0	0.0	45.3	0.194	46.0	tested...
14	2.0	68.0	62.0	13.0	15.0	20.1	0.257	23.0	tested...
15	2.0	68.0	70.0	32.0	66.0	25.0	0.187	25.0	tested...
16	10.0	68.0	106.0	23.0	49.0	35.5	0.285	47.0	tested...
17	1.0	71.0	48.0	18.0	76.0	20.4	0.323	22.0	tested...
18	1.0	71.0	62.0	0.0	0.0	21.8	0.416	26.0	tested...
19	2.0	71.0	70.0	27.0	0.0	28.0	0.586	22.0	tested...
20	1.0	71.0	78.0	50.0	45.0	33.2	0.422	21.0	tested...
21	9.0	72.0	78.0	25.0	0.0	31.6	0.28	38.0	tested...
22	1.0	73.0	50.0	10.0	0.0	23.0	0.248	21.0	tested...
23	5.0	73.0	60.0	0.0	0.0	26.8	0.268	27.0	tested...
24	0.0	74.0	52.0	10.0	36.0	27.8	0.269	22.0	tested...

Right click (or left+alt) for context menu

Add instance Undo OK Cancel

Se eliminan 5 instancias.

- **PASO 7:** Después de haber filtrado estos datos, volvemos a generar el árbol **J48** con el dataset depurado (ahora con 724 instancias), obteniendo el nuevo modelo.

=== Run information ===

Scheme: weka.classifiers.trees.J48 -C 0.25 -M 2
Relation: pima_diabetes
Instances: 724
Attributes: 9
preg
plas
pres
skin
insu
mass
pedi
age
class
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

J48 pruned tree

```
plas <= 143
| mass <= 26.3: tested_negative (131.0/3.0)
| mass > 26.3
| | age <= 28: tested_negative (210.0/34.0)
| | age > 28
| | | plas <= 99: tested_negative (51.0/8.0)
| | | plas > 99
| | | | pedi <= 0.527: tested_negative (107.0/44.0)
| | | | pedi > 0.527: tested_positive (54.0/15.0)
plas > 143: tested_positive (171.0/50.0)
```

Number of Leaves : 6

Size of the tree : 11

Time taken to build model: 0.03 seconds

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	528	72.9282 %
Incorrectly Classified Instances	196	27.0718 %
Kappa statistic	0.392	
Mean absolute error	0.3197	
Root mean squared error	0.4514	
Relative absolute error	70.8159 %	
Root relative squared error	95.0266 %	
Total Number of Instances	724	

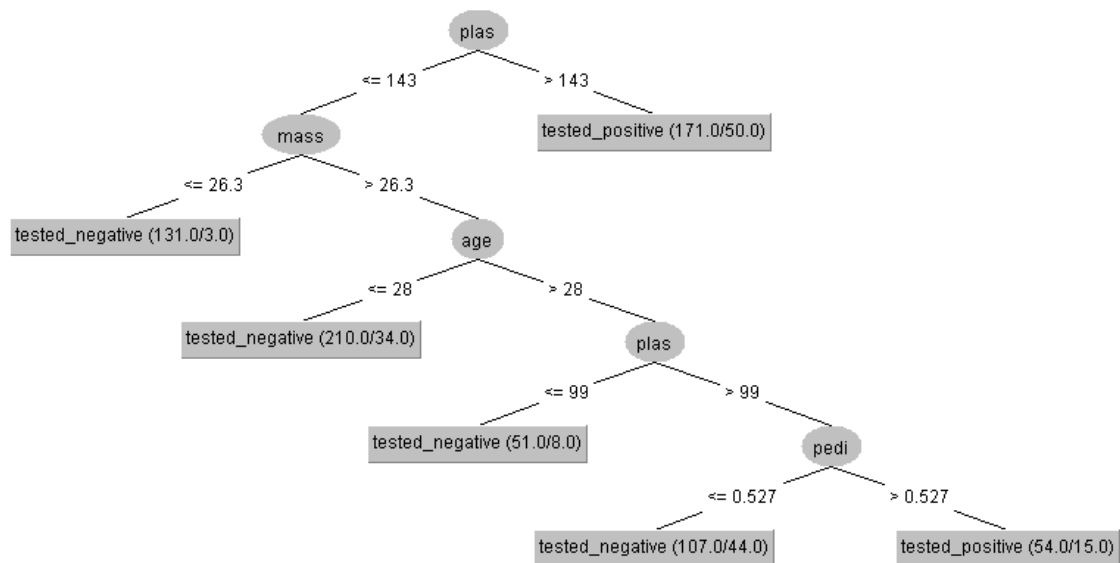
=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.808	0.422	0.785	0.808	0.797	0.392	0.737	0.801	tested_negative
	0.578	0.192	0.613	0.578	0.595	0.392	0.737	0.559	tested_positive
Weighted Avg.	0.729	0.343	0.726	0.729	0.727	0.392	0.737	0.717	

=== Confusion Matrix ===

```
a b <-- classified as
384 91 | a = tested_negative
105 144 | b = tested_positive
```

Nuevo árbol:



• **PASO 8:** Comparamos el rendimiento del modelo antes y después del filtrado en el siguiente cuadro:

Métrica	Antes (768 instancias)	Después (724 instancias)
Accuracy	73.83%	74.59%
Kappa	0.4164	0.4191
Tamaño del árbol	39 nodos, 20 hojas	11 nodos, 6 hojas
Falsos positivos	93	76
Recall (positivos)	0.597	0.566
RMSE	0.4463	0.4391

CONCLUSIONES

- El filtrado manual de valores imposibles (outliers) no mejoró drásticamente el *accuracy*, pero sí generó un árbol mucho más simple e interpretable, reduciendo el sobreajuste (*overfitting*) a datos ruidosos. Esto ocurrió a costa de una leve pérdida de sensibilidad (*recall*) para detectar la clase

positiva, posiblemente por la reducción en la cantidad de casos disponibles tras el filtrado.

- Es importante recordar que no se eliminaron las variables `insu` y `skin` a pesar de tener muchos ceros, porque al ser cantidades tan grandes, eliminarlas nos habría dejado con muy pocos datos para entrenar el modelo de manera efectiva.
- El cambio más significativo se dio en la complejidad del árbol: pasó de 39 nodos y 20 hojas a solo 11 nodos y 6 hojas. Esto indica que buena parte de la estructura del árbol original se dedicaba a "explicar" instancias ruidosas (datos faltantes mal codificados como 0) y no a capturar patrones reales de la enfermedad. Al depurar esos registros, el modelo resultante es considerablemente más claro y se apoya en solo cuatro atributos clave (`plas`, `mass`, `age`, `pedi`), dejando fuera `preg`, `pres`, `skin` e `insu` por su bajo poder predictivo en esta configuración limpia.